neighbors is smaller but the confidence interval does not cover 1. The total local effect of these licenses are smaller then when they were aggregated in previous models. The previous estimates of the local effect for all licenses was around 0.5 (for the linear predictor), and are 0.4, 0.3 and 0.1 for liquor, grocery and bars respectively, for an average effect of 0.27. In the case these components were orthogonal we would expect the average disaggregated effects added to be near the same size as the total effect, but in this sample they all have a slight positive correlation with one another so the reduction in effect sizes is not surprising (in addition to the other controls added into the model).

The neighbor effects for alcohol licenses are slightly smaller when disaggregating the different alcohol license types. In prior models the neighbor effect was around 0.22 in the linear predictor, and here are 0.2, 0.2 and 0.1 in the neighbor alcohol license effect for liquor stores, grocery stores and bars respectively. Again the neighboring effects appear to be *larger* than the local effects when considered how they diffuse to multiple adjacent regions. Given the standard errors of the local and neighbor effects the prior aggregation is not misleading. The effects are both in the same direction and in the same general order of size.

The average local effect of bars in this model is then around $\exp(0.3) \approx 1.3$, so if one added a bar to a location with an expected number of crimes being 1.5, the increase in crimes *on the local* street would be around half a crime, with the total expected to be close to 2 crimes. The increase in crimes on any *single* neighboring street is around $\exp(0.2) \approx 1.2$, and if the neighboring street has an expected number of 1.5 crimes the increase would be around 0.3 crimes for a total expected number of crimes being 1.8. Because this effect will diffuse to multiple streets though, typically around 6 to 8, this spatial diffusion will typically be larger combining all of the neighboring streets than the local effect. Going with a low estimate of the typical number of neighbors being 6, a typical increase in the total number of crimes when adding one liquor license will be slightly over 2 crimes combining both the local and the neighboring effect.[1]

The variables related to physical disorder, 311 calls for service, litter cans, toxic release sites, and vacant housing are all quite small and very close to 1 except for toxic release sites, which has a negative effect on crime. This is likely because the toxic release sites are industrial locations with little foot traffic. Although several of the coefficients for the 311 calls for service are estimated very precisely, they have nearly imperceivable effects on crime in this sample.

Measures of gentrification in this study (Wireless hot spots and coffee shops) have no obvious affect on crime; their confidence intervals cover 1. Some of the other variables that have been consistently found to affect crime, such as parks and bus stops, do not appear to affect crime in this cross-sectional sample. Bus stops are highly correlated with the inner parts of the city, and thus it is plausible the spatial trend terms removed the partial correlation in the subsequent model. The same could be said for coffee shops. The discretization of parks into multiple units may be partially responsible for the reason that parks are not correlated with crime when controlling for other factors. Shopping malls could be also be considered a proxy for gentrification, but they have a strong positive effect on crime. The generator effect of malls (e.g. they attract many of individuals) is surely much larger than any theoretical gentrification effect.

[1] As a robustness check to see if including intersections had any impact on the findings, I estimated interaction effects of intersections with local licenses, neighboring licenses, and local 311 calls for service (both Detritus and Infrastructure). All of the interactions were null *except* for intersection and neighboring licenses, which was positive. This is slight evidence that not including intersections slightly *decreases* the diffusion effects. This makes sense if intersections are more likely for crimes recorded outdoors, and so discarding intersections one would be discarding many crimes not at the local institution, but nearby.

Locations such as hospitals and police stations were included because these are often idiosyncratic locations where police record crimes as occurring. Hospitals are sometimes listed for assaults even if the assault occurred elsewhere and the victim was subsequently transported to the hospital. Police stations are similarly listed as the de facto location for certain crime complaints (e.g. if someone comes in person to file a report) although this occurs more often for minor crimes in the author's experience.

A street unit having a hospital on the premises is the largest effect in the model, with an increase in the incident rate ratio of over 4. The effect is quite imprecise though, and the confidence interval ranges from 1.3 to 14. Similarly the effect of a shopping center is quite large, with an incident rate ratio of over 2. Simply having a set of locations that predicts a large number of people in the area appear to be strong predictors of the number of crimes at small places.

10.3 Residual Diagnostics

As the same in prior presented models, the fit of the negative binomial model compared to the observed density in Figure 38 is quite good. There appears to be no need for zero inflated models, as the predicted number of zeroes given the negative binomial regression model fits the observed number of zeroes quite well. The tails of the distribution are predicted quite closely as well.

This section subsequently examines if any spatial autocorrelation still exists in the residuals of the model and the existence of outlier locations.

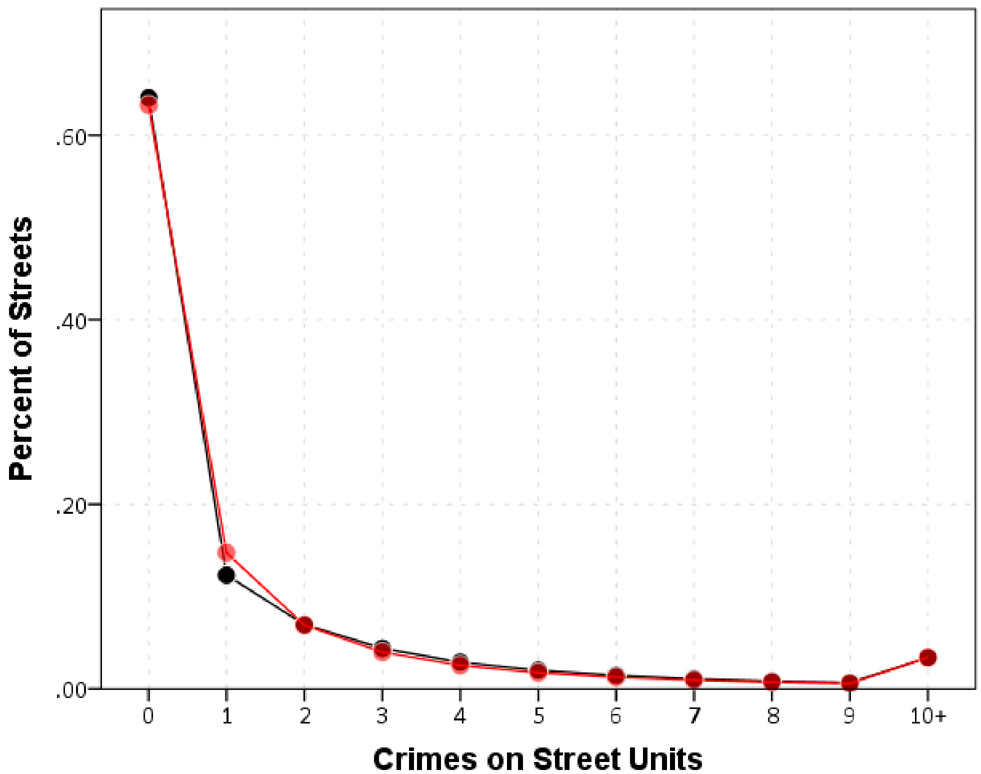


Figure 38: The fit of the negative binomial model is quite good. Only slightly under predicting zeroes and over predicting ones.

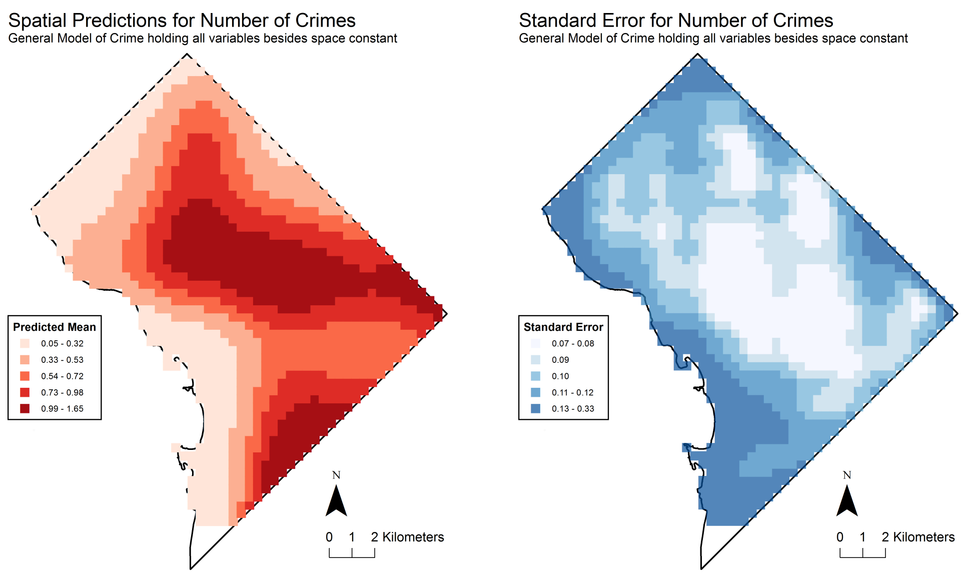
10.3.1 Residual auto-correlation

In the prior chapters, residual auto-correlation after fitting models was quite apparent, with Moran's I values between 0.05 and 0.15 depending on what types of residuals from the generalized linear model were examined. For this reason, fixed spatial terms were included in the model to detrend the data, and these terms consisted of restricted cubic splines of the projected x and y coordinates in meters as well as their interaction. This approach is more common in the natural sciences, but it has a nice analogue to the work of Shaw and McKay and the concentric rings model of crime emanating from the city center.

Figure 39 displays the ability of those terms to detrend the data. In the plot on the left are the mean predictions of crime over a regular grid holding covariates constant (for 0-1 or count covariates they are set to 0, for continuous covariates they are set to the mean). Comparing this to prior kernel density maps of crime shows a very similar pattern, and the terms are quite flexible enough to take the general trend of crime in particular areas of the city away. The map on the right displays the standard error of the predictions. The fixed spatial terms also have the advantage that they model the heteroscedasticity of the edge of the city explicitly.

Mapping the residuals of the model it is certainly an improvement over prior iterations. The residual map appears at first glance to be quite random. Presented in Figure 40 are the deviance residuals. Compared to previous maps, there are not strong clusters of under or over predictions in the model.

Examining the Moran's I value of the residuals using the same weights matrix for contiguity of the Thiessen polygons does reveal a slight amount of residual spatial auto-correlation



(a) (b)

Figure 39: These plots display the predicted mean of crime and the standard error of those predictions holding all variables in the model at constant values. All count and 0-1 variables are set to zero, and all area variables are set to their means. The predicted surface is then a function of the non-linear spatial restricted cubic spline terms, as is the predicted value holding these other covariates constant. This approximates the general trend of crimes across the city very well, and explicitly allows the data at the edge of the city to be heteroscedastic.

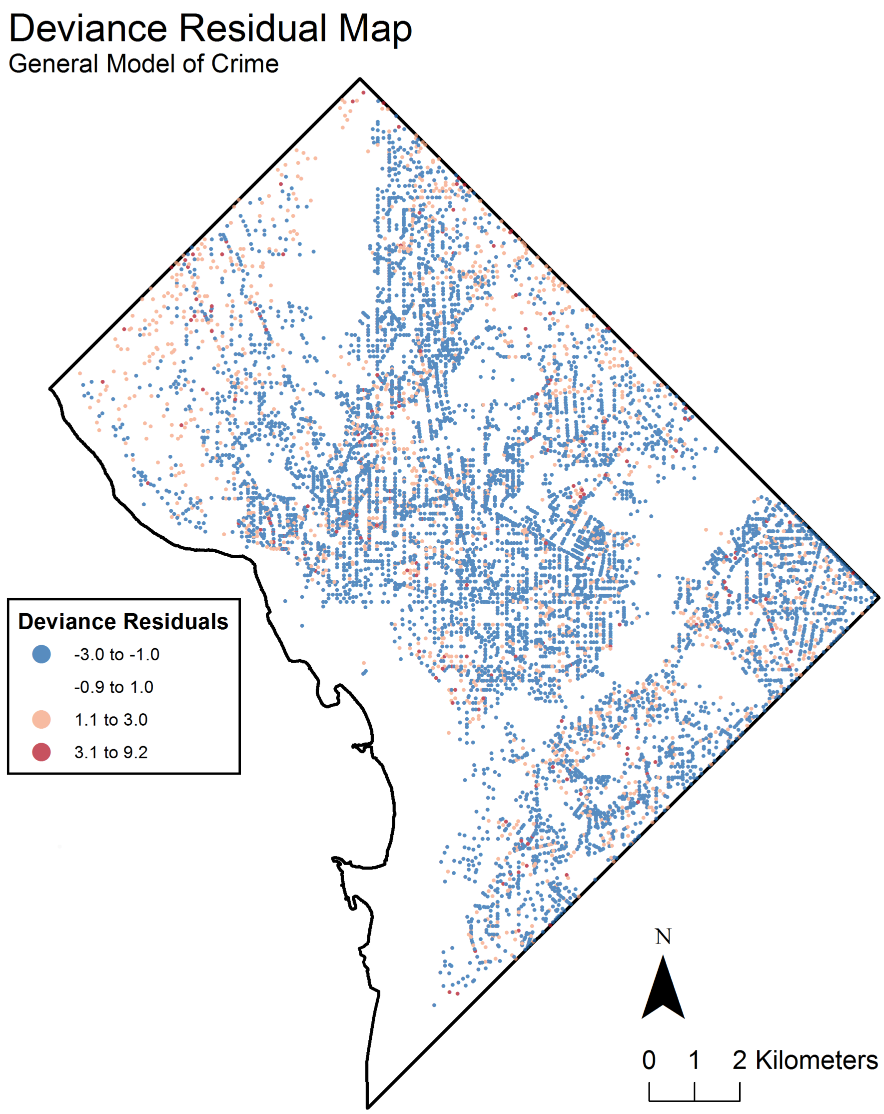


Figure 40: Deviance residuals for the general model of crime. As opposed to the prior models the residuals appear to be much more randomly distributed around the city.

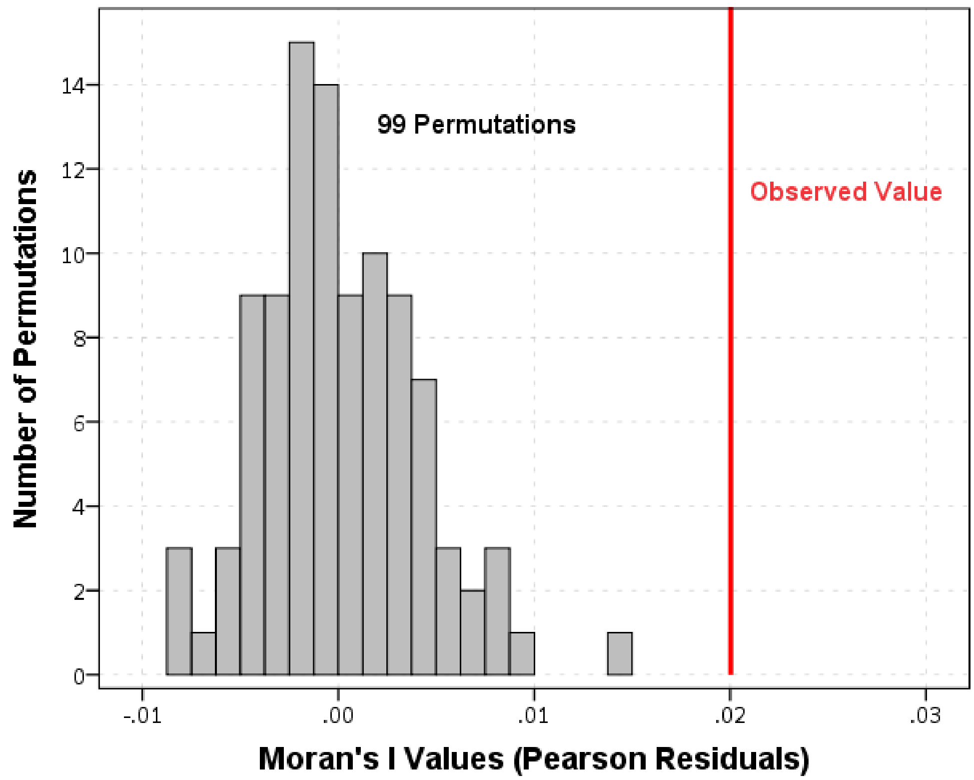


Figure 41: Moran's I values for Pearson residuals and for 99 permutations. The residual spatial autocorrelation is quite small, 0.02, but still significantly high compared to the null distribution.

still exists. Using the Pearson residuals and 99 permutations to generate a reference null distribution, a Moran's I value of 0.02 is observed, and the permutation distribution tends to be between -0.01 and 0.01, so the observed value is statistically significant at a 0.01 level. This test is displayed in Figure 41. Examination of the deviance residuals came to the same conclusion, and the observed value of Moran's I was slightly higher at 0.025.

Examining the correlogram at different distances, Figure 42 shows a similar story to that of the previous chapter. A negative auto-correlation is observed at very small distances of 100 meters or less, and then the auto-correlation swings back to positive and then generally declines for larger distances. Still the observed values are always outside of the 19 permutation grey lines until distance bins farther than $1,000$ meters.

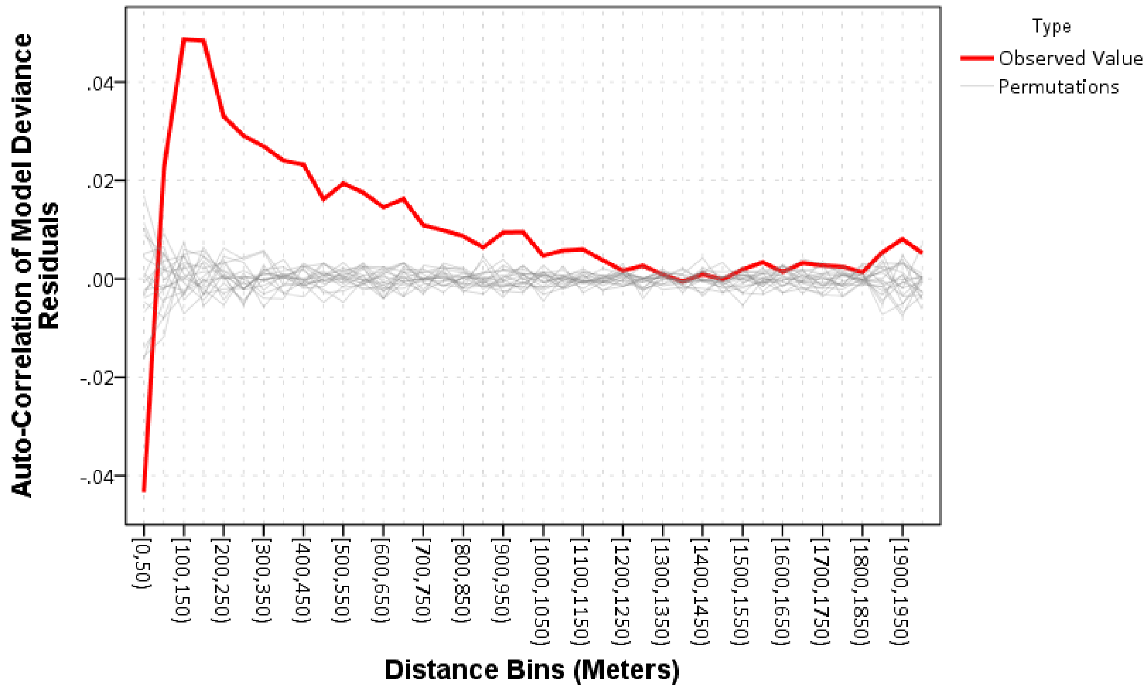


Figure 42: The spatial correlogram exhibits negative spatial auto-correlation at very short distances (under 100 meters) and then positive spatial auto-correlations at larger distance lags. The auto-correlations are smaller than the marginal crime totals, but are still unlikely to be generated by random spatial data having the same marginal distribution.

This suggests that the neighborhood differencing may not have been the culprit in the negative auto-correlation observed in the prior chapter. It is possible that the construction of the street units and the competitive process of assigning address coordinates is still not properly modelled. One may need to include spatial lags of the size of the neighbors as well as the size of the Thiessen polygon itself in the model. This correlogram also indicates that the spatial weights matrix only including contiguity neighbors is likely insufficient, as auto-correlation still exists at much higher distances of up to $1,000$ meters. The slow decay of the correlogram appears to be consistent with a moving average type spatial error.

10.3.2 Outliers

Comparing the outliers from the previous chapter in bars on crime, the model predicts the locations much more closely, but still tends to over predict the amount of crime at these particular locations. Again an online map is provided at `https://dl.dropbox.com/s/jxg2mjss6pvjam3/OutlierMap.html` where one can interactively see these 11 locations and their new predictions.

For the updated general model of crime seven new locations (and one of the prior locations which was greatly over-predicted) are included in a map of potential outlying locations. These locations were identified by examining model residuals, leverage and Cook's distance values. The current model did a much better job of predicting the higher crime locations over the previous iterations, and so including the wide set of generator locations likely increased the ability of the model to replicate reality. The spatial distribution of these outliers are dispersed throughout the city, unlike the prior model in which most were clustered in central D.C. Most of the outlier locations were now severely under predicted in their crime count. The findings are much more consistent with random outliers you might expect to find in 21,000 cases though, especially in terms of their spatial distribution.

10.4 Conclusion

This chapter presents an updated general model of crime at micro places. Many of the problematic aspects of the prior models, including poor predictions and spatial auto-correlation, were somewhat mitigated. The model still appears to have a slight amount of residual auto-correlation remaining, and the spatial correlogram shows that small distances still have a significant amount of negative auto-correlation and a positive auto-correlation at larger distances. This suggests that the overall auto-correlation may be the result of a mixture of negative and positive auto-correlation, and examination of the individual contributions to the global test statistic would be appropriate (Dray, 2011).

These model mis-specifications, as well as the potential for omitted variables, make interpreting the estimated effects in the model difficult. The different types of liquor licenses continue to be one of the larger effects on crime in the model, both locally and spatially. It is likely *some* of this effect is confounded with unobserved processes, such as liquor licenses self-selecting into locations that already have a larger amount of crime, but how much is unclear. A more appropriate research design might use other instruments to estimate the effect of bars on crime.

The effects of detritus and infrastructure 311 calls for service on crime were estimated as being statistically significant, but are incredibly small. These effects were similar in both models controlling for neighborhood effects as well as in the model presented in this chapter. The reason the effects are so precise is that calls for service are common throughout the city, so there is a great amount of variation. But, these effects appear inconsequentially small compared to other place based indicators of more people being in the area, such as a street unit being adjacent to a hospital or a public housing complex.

Other aspects of the built environment on crime have mixed results contrary to prior expectations. The null effect of parks is in particular surprising, and I suspect is partially a result of the conception of units of analysis. It is difficult to geocode locations within a park, and they are often assigned some arbitrary single location. With the discretization of street units, a park will be split up into multiple units, and so it may be that the one preferred geocoded location is high crime, but the rest of the locations are low crime.

Other measures of aspects of the built environment that should theoretically *decrease* crime, such as street lighting, trees, green sites, sidewalk cafes, and wireless hot spots, have either no discernible effect or slightly positive effects on crime. To appropriately identify the effect of these aspects of the built environment on crime will take more thought than a regression model that includes everything and the kitchen sink. Because these aspects of the built environment are very slowly changing, natural experiments are likely difficult to come by.

A unique contribution of the model presented in this chapter (in addition to the wide variety of aspects of the built environment) is the inclusion of non-linear spatial spline terms. While they do not greatly impact the estimated effects for most of the variables in the model, they do reduce the amount of spatial auto-correlation greatly as well as faithfully recreate the general trends in crime across the city. This provides a representation of crime that is a smooth function of space, as opposed to a neighborhood model of crime. A neighborhood model of crime implies that crime is monolithic within neighborhoods and discontinuous between neighborhoods, which is clearly inappropriate when examining the data.

Chapter 11

Conclusion and Future Goals

A recap of the empirical findings from the dissertation is quite simple:

- The number of liquor licenses at street units are a consistent predictor of Part 1 crimes. Adding one institution with a liquor license at a street unit will typically increase the yearly number of Part 1 crimes by around 2.

- Detritus and infrastructure 311 calls for service are associated with increased numbers of crimes on a street unit. The relationship is so small though to as be practically meaningless.

A hope of this dissertation though is to formalize *why* we want to use small units of analysis and *when* we **need** to use small units of analysis given the objectives of the study design. I was critical of prior work both suggesting one needed to use small units of analysis or that larger units of analysis were needed. The data analyses were vehicles to illustrate these points in applied examples.

Ultimately researchers in criminology will be interested in explaining not only why crime occurs, but also where crime occurs. I present a reductionist framework on why it is necessary to examine crime at small units of analysis, as otherwise we can not accurately represent or test our theories of crime at places. Simultaneously it also clarifies the role of aggregate level analysis, and how macro level analysis is not hopeless, just limited in the types of inferences one can make. It also has implications for how one interprets neighborhood level research, and how to *verify* a neighborhood level model one needs smaller units of analysis than the neighborhood.

11.1 Looking towards the Future

One of the grander claims in the dissertation is that crime at micro places do not conform to a neighborhood model of crime. That is, there appear to be no discrete boundaries where neighborhoods exist, and if one simply *assumes* neighborhood boundaries, one will find neighborhood effects even if there is a different data generating process. A spatial effects process as shown here in simulation and in empirical examples when aggregated up will look like a neighborhood effects model. Because of the prevalence of a neighborhood effects formulation, I demonstrate how one can examine between neighborhoods residual auto-correlation plots to determine if the data do not conform to the restrictive neighborhood model.

If one is concerned that a particular variable may be confounded with unobserved neighborhood effects I illustrated how using a fixed effects model can control for those omitted variables given you know the neighborhood boundaries. I did not however incorporate any demographic controls into the final analysis, such as the percentage of female headed households or the number of individuals below poverty at the micro place in the final model. It is demographically neutral. Other work has questioned the reasonableness of typical neighborhood boundaries (Hipp et al., 2011; Hipp and Boessen, 2013), and here I present a different formulation that accounts for general spatial trends by including non-linear spatial terms into the model. I believe the formulation I presented is a more realistic model of crime at micro-places, even absent these well established demographic predictors of crime. I suspect a similar view will not be held among many of my peers, as the relationship between demographic predictors of disadvantage and crime is so well established in macro level (Pratt and Cullen, 2005) and neighborhood level research (Sampson, 2012) it will appear strange not to include these factors in any model of crime.

There remains a large number of annoying methodological questions that need further investigation as to understand their impacts on the study of crime at micro places. While the prior chapters on how aggregation bias occurs shows why one wants to examine small units of analysis, this logic is unlikely to discriminate between different potential small units of analysis for many research questions, such as street segments and intersections used here or a fine grid such as is used in risk-terrain modelling (Kennedy et al., 2010). Using census boundaries with crime reports likely introduces serious error in assigning crime locations, as a large proportion of crime events will be geocoded to intersections that fall on the boundaries of census geographies. This problem is exacerbated if one is using crime already obfuscated to street midpoints, and in this case for the smallest census geography of a block nearly all crime falls on a border. For block groups this indeterminate amount is still incredibly large at around thirty percent in this sample. This makes the place for neighborhood covariates and crime even more uncertain, as allocation of demographic attributes to micro places will inevitably entail measurement error. Using street segments and intersections as a unit of analysis does not come without challenges though either, as these locations are not well defined for particular street layouts that occur in any non-perfect grid.

The number of potential covariates is immense as well. Prior work has examined land use, although different commercial establishments clearly show heterogeneity in the estimated effects presented here. Different types of liquor licenses (liquor stores, grocery and convenience stores, bars and restaurants) have approximately similar estimated effect sizes on crime in these models, but other commercial establishments are not as consistent predictors of crime. A street unit being adjacent to a hospital or a public housing complex predicts a fairly large increase in the rate of crime, and these types of institutions would not be typically captured if one was only modelling the generic land use classification. Simply obtaining all of these different predictors of crime in different study areas is quite a chore, although open data initiatives for many large cities make such data collection at least feasible.

In this analysis I limit the exploration of spatial effects to alcohol licenses and 311 calls for service. These spatial effects could easily be expanded to include all covariates of crime. It is a simple theoretical argument to suggest that these place based crime generators diffuse crime into nearby areas. It is also likely that the way that crime diffuses is not adequately represented in these models, and it may be more appropriate to consider places adjacent based on street connectivity than by adjacency of the Thiessen polygons (Groff, 2013).

Going forward while one may be able to reasonably predict crime at micro places, making inferences about the causes of crime is much more difficult. Aspects of the built environment are slowly changing, and make natural experiments to uniquely identify effects difficult. Here

I present some evidence that bars self-select into criminogenic locations by assuming that bars would have no effect on burglaries if there were no omitted variables in the model. As bars do have a slight effect on burglaries, it is likely the estimated total effect of bars on crime is over-stated. Other quasi-experimental designs will be needed to provide more reassurance that they observed effects are not spurious due to omitted variables.

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